```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 5, 11)
y = x**2 + 2

plt.plot(x, y, 'r')
plt.xlabel('X')
plt.ylabel('Y')
plt.title(r"$y=x^2+2$")

plt.savefig("plot.png", dpi=300, bbox_inches="tight")
```

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(0, 5, 11)
y = x ** 2+2
# plt.subplot(nrows, ncols, plot_number)
plt.subplot(2,1,1)
plt.plot(x, y, 'r--') # More on color options later

plt.subplot(2,1,2)
plt.plot(y, x, 'g*:')
plt.savefig('sub.png')
```

```
import matplotlib.pyplot as plt
import numpy as np

x = np.arange(0, 10, 0.1)
y = np.sin(x)

fig, ax = plt.subplots(figsize=(8, 4))

ax.plot(x, y)

ax.set_title("Sine wave")
ax.set_xlabel("x")
ax.set_ylabel("sin(x)")
ax.grid(True)

plt.savefig('sin.png')
```

Subplots

```
import matplotlib.pyplot as plt
import numpy as np

x = np.arange(0, 10, 0.1)
y = np.sin(x)

fig, ax = plt.subplots()

ax.plot(x, y, label="sin(x)")
ax.plot(x, np.cos(x), label="cos(x)")

ax.set_title("Trig functions")
ax.set_xlabel("x")
ax.set_ylabel("value")
ax.legend()
ax.grid(True)
plt.savefig("sub_trig.png")
```

Line plot

Scatter Plot

```


fig, ax = plt.subplots()

ax.scatter(x, y)

ax.set_title("Scatter plot")
ax.set_xlabel("x")
ax.set_ylabel("y")

plt.savefig('scatter.png')
```

Bar chart

```
categories = ["A", "B", "C"]
values = [10, 25, 15]

fig, ax = plt.subplots()

ax.bar(categories, values)
```

```
ax.set_title("Bar chart")
ax.set_xlabel("Category")
ax.set_ylabel("Value")

plt.savefig('bar.png')
```

Pandas and matplotlib

```
import pandas as pd

df = pd.DataFrame({
    "month": ["Jan", "Feb", "Mar"],
    "sales": [100, 150, 130]
})

fig, ax = plt.subplots()

ax.plot(df["month"], df["sales"], marker="o")

ax.set_title("Monthly sales")
ax.set_xlabel("Month")
ax.set_ylabel("Sales")

plt.savefig('summary.png')
```